Level 0x0c

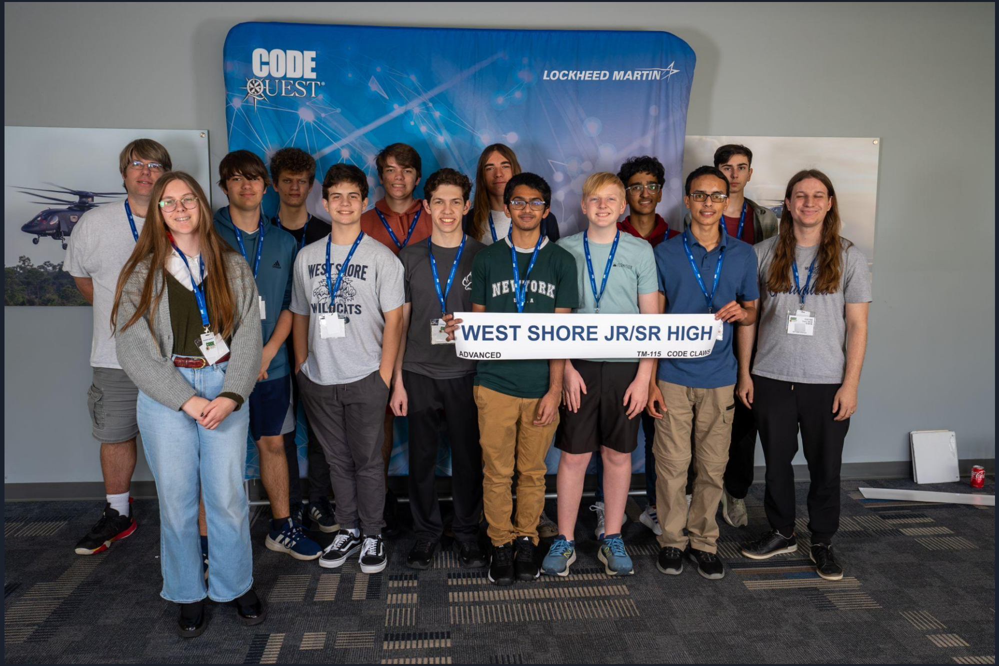
Engineering Degrees

Topics

- Events
- Code Quest

Cyber Quest and Code Quest

- Code Quest
    - Saturday, Feb 28th
    - Registration Nov 17th -
- Spring Break
    - March 23rd - 27th
- Cyber Quest
    - Saturday, March 28th
    - Registration Jan 5th -



Code Quest

LLM Ninjas Parker W Zachary W Charles L	Team 2 Last Names Nathan d W Ian B P Zunaira T
Wildcat New Blood Jasper B Kirin R Hope R	Wildcat Interns Sam S Eshan V Jay J

Code Quest

- Details
  - Saturday, February 28th, 9:00 - 1:30 ish
  - Teams are 2-3 students each, 4 teams max per school
  - 1 computer / laptop per team (one keyboard and one mouse too 😛)
  - No cell phones inside, no cameras (not even outside)
  - No buses, must provide your own transportation (they can’t stay on site during the competition)
  - Breakfast and Lunch provided
- What you need to provide me
  - Team members and teams (team names)
  - Birthday (must be 11 yrs and older, middle-school officially allowed)
  - Citizenship (not related to ICE current politics, standard procedure for military contractors)
  - Return 2 Lockheed forms (Liability and Photo Release)
  - Return 1 school permission slip

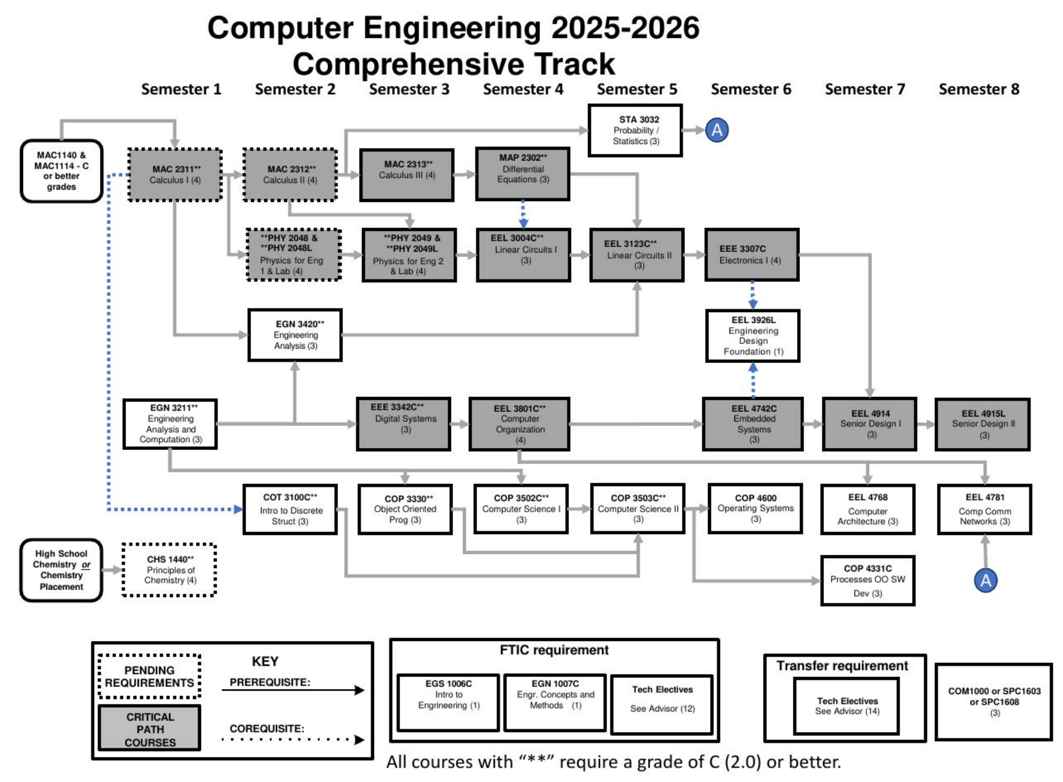
Cyber Quest

- Details:
    - Saturday, March 28th, 9:00 - 1:30 ish
    - Teams are 3-5 students each, 4 teams max per school
    - Very large monitors discouraged
    - No cell phones inside, no cameras (not even outside)
    - No buses, must provide your own transportation (they can't stay on site during the competition)
    - Breakfast and Lunch provided
- What you need to provide me
    - Team members and teams (team names)
    - Birthday (must be 14 yrs and older)
    - Citizenship (not related to ICE current politics, standard procedure for military contractors)
    - Return 2 Lockheed forms (Liability and Photo Release)
    - Return 1 school permission slip

Undergraduate Catalog

- University Education in constant flux
  - New technologies replace older technologies
  - Engineering and Computer Science especially
- Graduation requirements outlined in Undergrad Catalog
  - Published every year / subject to change
  - You get locked into catalog for your enrollment year
  - You can change to newer catalog, but never backwards
- Buy physical or download PDF of your Undergraduate Catalog
  - Know the requirements of your major
  - Talk to your department
  - Be a little wary of general academic advisors

UCF Computer Engineering



Catalog Pages

General Education (GEP)

Complete all of the following
The UCF General Education Program (GEP) is described in this catalog. Engineering students should closely study the requirements of the UCF GEP and the allowable substitutions detailed in paragraphs A through E below to minimize excess hours. Students transferring to UCF from within the Florida College System or State University System should complete the GEP and the Common Program Prerequisites before transferring.
Communication Foundations
Complete all of the following
Required:
Complete the following:
ENC1101 - Composition I (3)
ENC1102 - Composition II (3)
Suggested:

Complete at least 1 of the following:
SPC1603C - Fundamentals of Technical Presentations (3)
SPC1608 - Fundamentals of Oral Communication (3)
Cultural and Historical Foundations
Complete all of the following
Earn at least 6 credits from the following types of courses:
Historical Foundations
Earn at least 3 credits from the following types of courses:
Cultural Foundations
Mathematical Foundations
Complete all of the following
Required:
Complete the following:
MAC2311C - Calculus with Analytic Geometry I (4)
STA3032 - Probability and Statistics for Engineers (3)

Social Foundations
Complete all of the following
Preferred:
Complete the following:
ECO2013 - Principles of Macroeconomics (3)
ECO2023 - Principles of Microeconomics (3)
Or
Select one class from Social Foundations Credit Hours: 3
Science Foundations
Complete all of the following
Required:
Earn at least 4 credits from the following course sets:
keyboard_arrow_up
PHY 2048C
PHY2048C - General Physics Using Calculus I Studio (4)
keyboard_arrow_up
PHY 2048 and 2048L
PHY2048 - General Physics Using Calculus I (3)
PHY2048L - General Physics Using Calculus I Laboratory (1)
Earn at least 3 credits from the following course sets:
keyboard_arrow_up
GEP 11- Science Foundation
AST2002 - Astronomy (3)
CHM1020 - Concepts in Chemistry (3)
CHM1032 - General Chemistry (3)
CHM2045C - Chemistry Fundamentals I (4)
CHS1440C - Principles of Chemistry (4)
PHY1038 - Physics of Energy, Climate Change and Environment (3)
PHY2020 - Concepts of Physics (3)
PHY2048 - General Physics Using Calculus I (3)
PHY2053 - College Physics I (3)

Common Program Prerequisites (CPP) (If applicable)

Complete all of the following
These courses are specifically required for all engineering students of the Florida State University System. CPP courses are also available at other Florida post-secondary schools and may be transferred directly to UCF programs. To enroll in CpE major courses, a 2.0 (C or better) in each course is required.
See “Common Prerequisites” in the Transfer and Transitions Services section for more information.
Earn at least 15 credits from the following:
MAC2311C - Calculus with Analytic Geometry I (4)
MAC2312 - Calculus with Analytic Geometry II (4)
MAC2313 - Calculus with Analytic Geometry III (4)
MAP2302 - Ordinary Differential Equations I (3)
PHY 2048C - General Physics Using Calculus I (or PHY2048 and PHY2048L) Credit Hours: 4 (GEP)
PHY 2049C - General Physics Using Calculus II (or PHY2049 and PHY2049L) Credit Hours: 4
GEP Courses: MAP 2311C, PHY 2048C/PHY2048 & PHY 2048L
Complete at least 1 of the following:
CHS1440C - Principles of Chemistry (4)
CHM2045C - Chemistry Fundamentals I (4)
* Preferred: CHS 1440

Grand Total Credits: **19**

Computer Engineering

- General Education (57 hrs)
  - 38 hr general education
  - 19 hrs common pre-reqs
- Degree Requirements (71 hrs)
  - 51 hrs required CS/CpE
  - 12 hrs tech electives
  - 6 hrs senior design
- 128 credit hours

Courses Required for the Major
51 Total Credits
keyboard_arrow_up

Complete all of the following
Complete the following:

COT3100C - Introduction to Discrete Structures (3)
COP3502C - Computer Science I (3)
COP3503C - Computer Science II (3)
COP3330 - Object Oriented Programming (3)
COP4331C - Processes for Object-Oriented Software Development (3)
EEL3926L - ECE Design Lab (1)
EEL3004C - Linear Circuits I (3)
EEL3123C - Linear Circuits II (3)
EEE3307C - Electronics I (4)
EEE3342C - Digital Systems (3)
EEL3801C - Computer Organization (4)
EEL4742C - Embedded Systems (3)
EEL4768 - Computer Architecture (3)
EEL4781 - Computer Communication Networks (3)
COP4600 - Operating Systems (3)

EGN3211 - Engineering Analysis and Computation (3)
EGN3420 - Engineering Analysis (3)

A "C" (2.0) or better is required in the following courses: COP 3330, COP 3502C, COP 3503C, COT 3100C, EEE 3342C, EEL 3004C, EEL 3123C, EEL 3801C, EGN 3211, EGN 3420

Restricted Electives
12 Total Credits
keyboard_arrow_up

Technical electives are available in the BSCpE program to address specific student interests in a variety of technical areas such as Software Engineering. Students should consult with their academic advisor for the identification of courses that are approved technical electives and the terms when specific courses of

Capstone Requirements
6 Total Credits
keyboard_arrow_up

Complete the following:

EEL4914 - Senior Design I (3)
EEL4915L - Senior Design II (3)

Grand Total Credits: **71**

Computer Science

- General Education (56 hrs)
  - 42 hr general education
  - 14 hrs common pre-reqs
- Degree Requirements (60 hrs)
  - 30 hrs required CS
  - 18 hrs tech electives
  - 6 hrs advanced math
  - 6 hrs senior design
- 120 credit hours

Degree Requirements

Core Requirements: Basic Level
30 Total Credits
keyboard_arrow_up

Complete all of the following
Earn at least 27 credits from the following:

STA2023 - Statistical Methods I (3)
COP3330 - Object Oriented Programming (3)
COP3502C - Computer Science I (3)
COP3503C - Computer Science II (3)
CDA3103C - Computer Logic and Organization (3)
COT3100C - Introduction to Discrete Structures (3)
CIS3360 - Security in Computing (3)
COP3402 - Systems Software (3)
COT4210 - Discrete Structures II (3)
COP4331C - Processes for Object-Oriented Software Development (3)
COT3960 - Foundation Exam

Complete at least 1 of the following:

ENC3241 - Writing for the Technical Professional (3)
ENC3250 - Professional Writing (3)

Grand Total Credits: **30**

Program Details

Core Requirements: Advanced Level (18 Credit Hours)

Cybersecurity Area

CAP 4145 - Introduction to Malware Analysis **Credit Hours: 3**
CIS 3362 - Cryptography and Information Security **Credit Hours: 3**
CIS 4203C - Digital Forensics **Credit Hours: 3**
CIS 4361 - Secure Operating Systems and Administration **Credit Hours: 3**
CIS 4615 - Secure Software Development and Assurance **Credit Hours: 3**
CIS 4940C - Topics in Cybersecurity **Credit Hours: 3**
CNT 4403 - Network Security and Privacy **Credit Hours: 3**
EEE 4346C - Hardware Security and Trusted Circuit Design **Credit Hours: 3**

AI and Machine Learning Area

CAP 4053 - AI for Game Programming **Credit Hours: 3**
CAP 4453 - Robot Vision **Credit Hours: 3**
CAP 4611 - Algorithms for Machine Learning **Credit Hours: 3**
CAP 4630 - Artificial Intelligence **Credit Hours: 3**
CAP 5415 - Computer Vision **Credit Hours: 3**
CAP 5512 - Evolutionary Computation **Credit Hours: 3**
CAP 5610 - Machine Learning **Credit Hours: 3**
CAP 5636 - Advanced Artificial Intelligence **Credit Hours: 3**

UF Computer Engineering

Degree Reqs (56 hrs total)

- 38 hrs CpE Reqs
- 12 hrs Tech Electives
- 6 hrs Senior Design

126 total credit hrs required

Computer Engineering Requirements

A minimum grade of C is required for each critical-tracking course and the critical-tracking GPA must be a minimum of 2.5.

A minimum grade of C is required in any computer engineering course that is a prerequisite for another computer engineering course and CpE Design 2 CEN 4908C. The prerequisite course and its subsequent course cannot be taken the same term, even if the prerequisite course is being repeated.

Minimum grades of C are required in:

Code	Title	Credits
CDA 3101	Introduction to Computer Organization	3
CEN 3031	Introduction to Software Engineering	3
COP 3502C	Programming Fundamentals 1	4
COP 3503C	Programming Fundamentals 2	4
COP 3504C	Advanced Programming Fundamentals for CIS Majors	4
COP 3530	Data Structures and Algorithm	3
COT 3100	Applications of Discrete Structures	3
EEL 3701C	Digital Logic and Computer Systems	4
EEL 4744C	Microprocessor Applications	4
ENC 3246	Professional Communication for Engineers	3
CpE Design 1		
Select one:		3
CEN 3907C	Computer Engineering Design 1	
EGN 4951	Integrated Product and Process Design 1	
CpE Design 2		
Select one:		3
CEN 4908C	Computer Engineering Design 2	
EGN 4952	Integrated Product and Process Design 2	

Electrical Engineering

- 6 classes changed from computer engineering

- These are all intensely challenging

Courses Required for the Major
47 Total Credits
keyboard_arrow_up

Complete all of the following

Complete the following:

EEL3926L - ECE Design Lab (1)
EGN3211 - Engineering Analysis and Computation (3)
EEL3004C - Linear Circuits I (3)
EEL3123C - Linear Circuits II (3)
EEE3350 - Semiconductor Devices (3)
EEE3307C - Electronics I (4)
EEE3342C - Digital Systems (3)
EEL3470 - Electromagnetic Fields (3)
EEL3552C - Signal Analysis and Analog Communication (4)
EEL3657 - Linear Control Systems (3)
EEL3801C - Computer Organization (4)
EEE4309C - Electronics II (4)
EEL4750 - Digital Signal Processing Fundamentals (3)
EEL4742C - Embedded Systems (3)
EGN3420 - Engineering Analysis (3)

A "C" (2.0) or better is required in the following courses: EEE 3342C, EEL 3004C, EEL 3123C, EEL 3801C, EGN 3211, EGN 3420

Restricted Electives
16 Total Credits
keyboard_arrow_up

Technical electives are available in the BSEE program to address specific student interests in a variety of technical areas. Students should consult with their academic advisor for the identification of courses that are approved technical electives and the terms when specific courses of this type are to be offered.

Links